

Charan Mogulla

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SUMMARY

AI Engineer with about 3 years of ML experience, including roughly 2 years working on LLM applications. Most of that recent work has been retrieval systems, fine-tuning open-source models with LoRA and QLoRA, and shipping the resulting services on Azure. Earlier background is in Kafka and Spark data pipelines.

SKILLS

- **LLM and GenAI:** RAG, prompt engineering, LoRA, QLoRA, LangChain, OpenAI, Azure OpenAI, embeddings
- **ML and Deep Learning:** PyTorch, Hugging Face Transformers, scikit-learn
- **Vector search:** FAISS, Azure AI Search
- **Data and streaming:** Apache Kafka, Apache Spark, Spark Streaming, SQL, ETL
- **MLOps:** Docker, Kubernetes (AKS), Azure DevOps, Azure Monitor, MLflow
- **Cloud:** Azure (ML, OpenAI, Functions, App Services, AKS)
- **Backend:** FastAPI, Flask, REST APIs
- **Languages:** Python, SQL

EXPERIENCE

Staffingtree Inc.

Aug 2023 - Present

AI Engineer

Overland Park, KS

- Built a RAG document Q&A system using LangChain, OpenAI embeddings, and FAISS, with a FastAPI backend. Median response time on production traffic sits a little under 2 seconds for typical queries.
- Wrote the ingestion and chunking pipeline for PDF, HTML, and Markdown sources. Tuned chunk size and overlap by running retrieval evals against a labeled question set - the shipped version had roughly 30% fewer retrieval misses than the first pass.
- Fine-tuned Mistral 7B and LLaMA-2 with LoRA and QLoRA on a domain Q&A dataset, then moved the high-volume path off GPT-4 onto the tuned models. Per-query inference cost dropped by around 40% with comparable answer quality on our eval set.
- Set up an evaluation harness with automated prompt benchmarks plus a small reviewer queue. Prompt and model changes have to pass it before they can go out.
- Added a multi-document agent flow with LangChain tools and Azure Functions for queries that need to pull from more than one source in a single response.

Infosys

Jul 2022 - Jul 2023

Systems Engineer (ML Engineer track)

Hyderabad, India

- Prototyped a knowledge assistant on Azure AI Search using vector retrieval; moved from internal prototype to pilot before role end.
- Built Azure ML pipelines for ingestion, training, eval tracking, and deployment. Release cycles for new model versions went from about three weeks down to roughly one.
- Packaged FastAPI microservices in Docker on AKS through Azure DevOps. Versioning was set up so a rollback was one command instead of a redeploy.
- Hooked retrieval-based answer suggestions into the call center workflow for a few intents and saw measurable handle time reduction on the tracked subset.

HCL Technologies

Dec 2020 - Jun 2022

Data Engineer

Hyderabad, India

- Built and maintained Kafka and Spark Streaming pipelines for real-time logistics and transaction events that fed downstream fraud detection and analytics.
- Cut fraud detection latency from a few minutes down to under 30 seconds by reworking the ingestion path and tuning Spark Streaming micro-batch intervals.
- Wrote Spark SQL ETL jobs with schema validation, deduplication, and quality checks. Cut down on downstream pipeline failures caused by upstream schema drift.

PROJECTS

Checkmate - AI Code Review GitHub App

- Reviews pull requests with Claude and posts inline comments scoped to the changed diff lines. Indexes the target repo on first use into Qdrant with BGE-small embeddings so the reviewer sees function-level context, not just the diff. Langfuse traces every call (tokens, latency, cost), and a promptfoo suite of 14 test cases has to pass before any prompt change ships. FastAPI service with an RQ worker queue, Postgres for state. Code on GitHub.

Domain-Specific LLM Fine-Tuning

- Mistral 7B with QLoRA on a curated instruction dataset I assembled (~8K pairs). Compared the tuned model against the base model on a held-out eval set; tracked exact-match and a small human-rated quality score. Code and write-up on GitHub.

EDUCATION

University of Central Missouri	Aug 2023 - May 2025
M.S., Computer Science (AI & ML)	
• GPA: 3.6 / 4.0	
Bharat Institute of Engineering and Technology	Jun 2017 - Jul 2021
B.Tech, Computer Science	
• GPA: 8.3 / 10	

CERTIFICATIONS

• DeepLearning.AI: Generative AI with LLMs · Deep Learning Specialization · LangChain for LLM Application Development
